Unit: 6 Unit 4: Absolute Value **Name:** ________________

Topic: Absolute Value Function and its Graph **Date:** ____________

Foundation (Jalapeno)

Q1. What is the graph of $y = |x|$?
(A) A straight line
(B) A parabola
(C) A V-shaped graph
(D) A circle
(E) An ellipse

Q2. What is the vertex of the graph of $y = |x - 2|$?
(A) $(0, 0)$
(B) $(2, 0)$
(C) $(2, 2)$
(D) $(-2, 0)$
(E) $(0, 2)$

Q3. What is the axis of symmetry of the graph of $y = |x + 1|$?
(A) $x = -1$
(B) $x = 0$
(C) $x = 1$
(D) $y = -1$
(E) $y = 0$

Q4. What is the direction of opening of the graph of $y = |x|$?
(A) Up
(B) Down
(C) Left
(D) Right
(E) None

Q5. What is the graph of $y = 2|x|$?
(A) A narrow V-shaped graph
(B) A wide V-shaped graph
(C) A straight line
(D) A parabola
(E) A circle

Q6. What is the graph of $y = |x| - 1$?
(A) A V-shaped graph shifted up
(B) A V-shaped graph shifted down
(C) A V-shaped graph shifted left
(D) A V-shaped graph shifted right
(E) A straight line

Q7. What is the graph of $y = |x + 2| + 1$?
(A) A V-shaped graph shifted up and left
(B) A V-shaped graph shifted up and right
(C) A V-shaped graph shifted down and left
(D) A V-shaped graph shifted down and right
(E) A straight line

Q8. What is the vertex of the graph of $y = |x - 3| + 2$?
(A) $(3, 0)$
(B) $(3, 2)$
(C) $(0, 2)$
(D) $(0, 3)$
(E) $(2, 3)$

Open Ended Questions (FRQ)

Q9. Graph the function $y = |x - 1| - 2$ and identify its vertex, axis of symmetry, and direction of opening.

Q10. Compare and contrast the graphs of $y = |x|$ and $y = |x| + 1$. How do the graphs differ?

Chef's Tips

- Tip 1: Review the definition of absolute value and its graph
- Tip 2: Emphasize the importance of identifying the vertex, axis of symmetry, and direction of opening
- Tip 3: Encourage students to use graphing tools to visualize the transformations